

Study Title: *The influence of response effector on decision adaptations*

Information leaflet

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Thank you for considering participating in this study of how our decisions are intertwined with our movements. This study is funded by **Science Foundation Ireland**. Before agreeing to participate, please take your time to read through this information leaflet and ask any questions you have, and do not feel any pressure.

What is this research about?

During everyday life, you are continuously required to make decisions about your sensory environment. We, however, still know surprisingly little about how our brain translates what we see or hear into a decision. The aim of this study is to investigate how people collect sensory evidence to make a decision, how this depends on the actions we are performing, and to examine the underlying neural mechanisms.

Why are we doing this research?

This research will contribute to a better understanding of mechanisms of perceptual decision-making, which are fundamental to general brain function. Although for now we are pursuing this knowledge for its own sake, our findings may be useful to future research aimed at developing diagnostics and clinical tools for psychiatric and neurological disorders.

Why have you been invited to take part?

You have been invited to take part in this experiment because you are an adult (aged 18-50) with normal or corrected-to-normal vision, and have no diagnosis of a neurological or psychiatric disorder that could significantly impair your ability to produce immediate actions based on sensory stimuli. Since some stimuli flash on and off, to be conservative, we ask you not to participate if you have any **sensitivity to flickering light**, a personal history of **unexplained fainting**, or history of **epilepsy**. Because the experiment involves placing silver EMG electrode contacts on the skin of your hands/arms, you cannot participate in EMG experiments if you have a **silver allergy**. In addition, because of a very low risk of malfunction, you may not participate if you have **implanted electronic devices of any kind**, including cardiac pace-makers or similar assistive devices, electronic infusion pumps, and implanted stimulators.

What will happen if you decide to take part in this research study?

If you decide to take part we will ask you to sign the consent form below, and to come in to the human neurophysiology lab in the engineering building (Room 022) for **1** recording sessions of approximately **3.5** hours each. When you arrive the researcher will offer to wear a mask, and one will be provided to you, if it would make you more comfortable to minimize the risk of COVID-19 transmission.

We will ask you to complete a short demographic form with the following information: Age, gender (if you choose to provide it), and handedness. We will also measure the dimensions of the muscles from which we will be recording (electromyography, or EMG) and record these on this form.

In the experiment you will sit in front of a computer screen and watch simple stimuli at normal intensity levels such as moving dots, striped patterns or coloured shapes, and make a judgment on a simple attribute of the stimulus like direction, orientation, colour, etc. You will report your decision either by clicking a button or shifting your gaze to a target, or by activating your muscles of your finger and thumb which we will measure using electromyography (EMG). For the EMG-based response, you will learn in advance how to activate your muscles to the required levels by viewing the measured level of muscle activation on the screen. You will do this repeatedly in blocks of **10** minutes, resting for as long as you like in between. While performing the task you will be in a closed dark booth but during rest breaks you will be able to chat to the researcher, take refreshments, etc. We ask you to perform **10** such blocks in a session, but if you become tired or for any other reason wish to stop, you can do so at any time. The instructions at the beginning of the task will tell you exactly what the task demands are - what are the possible stimuli and what movements are the correct ones for each type, how quickly to respond, the level of muscle activation needed, and any other information such as how difficult you can expect it to be. To help make it fun, and so “good performance” is clearly defined, we assign different numbers of points to different outcomes (e.g. correct, on-time, incorrect, late) and give you feedback, much like in a computer game, so you can continually try to improve your score. If the task is tricky, we may ask you to come to the lab for one or two short (20-40 min) training session on a day in advance of the EEG recordings.

In the EEG recording session we will place an EEG cap on your head. This is a flexible cloth cap, much like a swimming cap, with holes that hold the electrodes. These electrodes are non-invasive and will sit on top of your scalp, picking up from afar the electrical activity produced by your brain as you process the stimuli. To record this activity we squirt a little conductive gel in each hole onto your scalp using a blunt plastic syringe. This syringe does not pierce or damage your skin, but might possibly feel a little uncomfortable. We will take care during the set up to minimize any such discomfort but let us know. The gel is non-toxic and washes out easily with water. It feels a bit cold when it is first applied but this feeling passes quickly. Small additional electrodes will be taped to the surface of your skin on your hands, and other similar electrodes on your cheek and forehead, using soft, easily removable, hygienic surgical tape.

While you are engaged in the experiment, we will ask you to keep still, and while deciding about the stimulus, to keep your eyes fixated on the screen. A camera-based, remote eye tracker on the desk in front of you will be recording your eye movements while you perform the task.

How will your data be used ?

For this experiment each participant will be assigned a code by which no one but the researchers named on this study will be able to identify you, and your data will be stored using only this code. This is called pseudonymising the data. The key to the code remains with the researchers, separate from any of the data, and will be deleted after data collection and some preliminary analyses are complete for this experiment.

When we have collected all the EEG, EMG, behaviour and eye-tracking data, we will publish the results of the full group of subjects together, in which it will not be possible to identify you by your data. With your permission, your pseudonymised data will be retained for possible future analyses and may be shared with other scientists who wish to gain further insights from them, by uploading the data to an online database to which anyone can access. The demographic information will be included with this data, but no information about your identity will ever be shared with other researchers.

How will your privacy be protected?

As stated above, we will store, process and (with permission) share your data only in pseudonymised form. The demographic information will be entered into a google spreadsheet kept on UCD servers and will contain your code, not your name.

After participation, we will delete your contact information by default. However, in case you are interested in being informed of future opportunities to participate in behavioural and EEG research of the same nature, we have included a section below the consent form allowing you to indicate whether you give permission to retain your email address for this purpose. If you give this permission to be contacted again, we will store your email address separately from the data that we collected in a password-protected file maintained on UCD Connect. This file will only be accessible by the researchers associated with the Cognitive Neural Systems Lab. You can change your mind and withdraw your contact information at ANY stage, and you do not need to give a reason for withdrawing. If/when we email you with new study information, we will always attach an information letter relevant to that new study. You will be asked for permission again with every new participation. If we do not get further instruction from you, your contact information will be removed from the list after 5 years.

What are the benefits of taking part in this research study?

There are no direct advantages and disadvantages associated with this study. The research will provide valuable insights into the underlying neural mechanisms of perceptual decision-making and this may prove useful in future work aimed at understanding and/or diagnosing psychological and neurological disorders.

If you happen to be a student of one of the researchers, please note that participation in this experiment has no relation whatsoever to your academic standing.

What are the risks of taking part in this research study?

You might experience some discomfort or fatigue as you spend a prolonged period sitting still and concentrating on a computer screen. You will be given regular breaks to minimize this discomfort.

The conductive gel can be messy and may stick to clothes/hair, but it washes out easily with water.

None of the methods we carry out in this study will relate to medical diagnosis.

The EEG set-up is very similar in nature, level and duration of contact to that of hairdressers, and therefore involves the risk of spreading COVID-19. Masks will be available in the laboratory to be worn. Depending on the public health conditions of the time, the researcher may discuss the use of masks with you over email before scheduling the session, to make sure both of you are comfortable.

Can you change your mind at any stage and withdraw from the study?

Yes. You decide whether to participate in the study. Participation is voluntary. If you decide not to participate, please let us know. You do not have to sign anything you do not wish to. If you do take part, you can change your mind and withdraw at ANY stage of the experiment, up until the data collection is completed and the key linking your name to the data has been deleted. You do not need to give a reason for withdrawing.

How will you find out what happens with this project?

If you would like to find out more about the results of this study, you can contact any of the investigators listed in this information sheet and we will be happy to provide this information to you. Contact information is listed at the bottom of this information sheet.

Contact information:

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